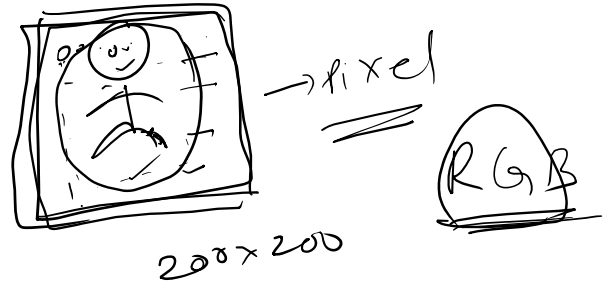


numpy ✓① Scalarimport numpy as np -

a=10, b=20.5
simple

a = np.array(20)

a = 20

a.ndim → 0

a.shape ()② vector -

a_vector = np.array([1, 2, 3, 4, 5])

a_vector → [1 2 3 4 5] -

a_vector.ndim → 1

a_vector.shape → (5,)

Matrix

A = $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3}$

(20)

[]

[]

Y

X

$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

3) Matrix

a_matrix = np.array([[1, 2, 3],
[4, 5, 6],
[7, 8, 9]])

Scalar $\rightarrow$ no dimension

4) tensor $\rightarrow$ Google

vector $\rightarrow$ 1

matrix 2

tensor n dim

array () ✓

([])

([[]])

[[[]]]

a-tensor = np.array (

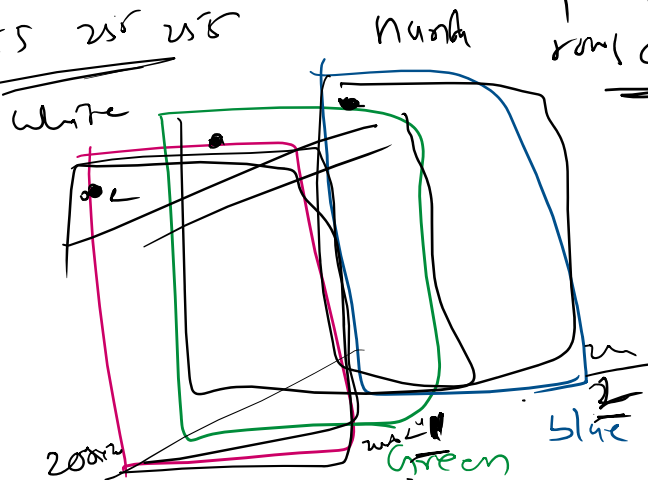
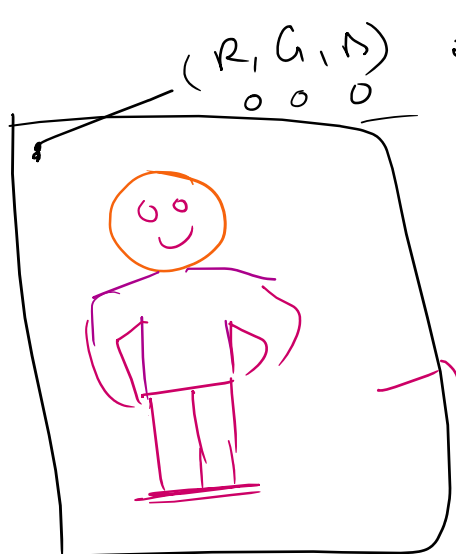
$\left[\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 10 & 11 & 12 \\ 13 & 14 & 15 \\ 16 & 17 & 18 \end{pmatrix} \right]$

tensor

~~3x3~~

2 x 1 x 3

num row col



R, G, B

200x200

2 $\rightarrow$ [[[]]]

whole video 4D = (num-frames, height, width, dim)
(2048 640 500 3)

SD = (width, num-h, len, any dim)

2D - Images (rows, cols)

3D - single img (height, w, channel)

UD $\rightarrow$ (num, h, w, (head))